Informatique et systèmes de communication ISC

Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

Signal and information (SigInf)

204.1

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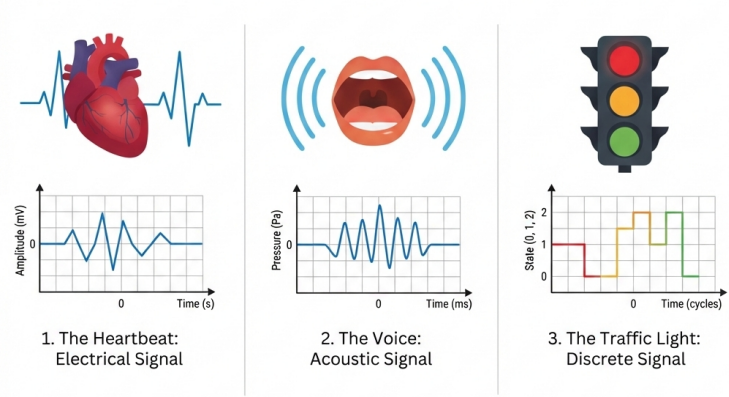
23.2.2026

Structure

- Definition of a signal (continuous and discrete)
- Operations on signals
- Energy and power
- Types of signals
- Noise
- Sampling

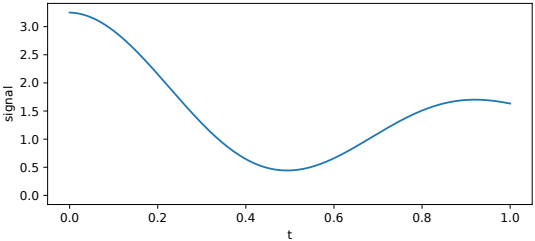
P. Prandoni and M. Vetterli, *Signal processing for communications*, EPFL Press (2008).

Examples


Amplitude (mV)
0
0
Time (s)
1. The Heartbeat:
Electrical Signal
Pressure (Pa)
0
0
Time (ms)
2. The Voice:
Acoustic Signal
State (0, 1, 2)
2
1
0
0
Time (cycles)
3. The Traffic Light:
Discrete Signal

What is a signal?

A signal is a physical quantity that varies over time, space, or another independent variable, and it conveys information.



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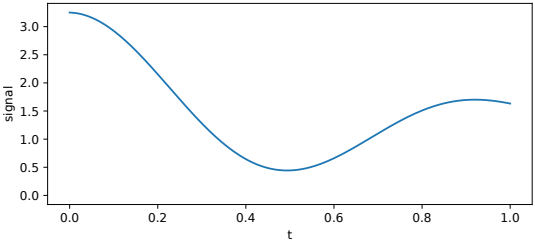

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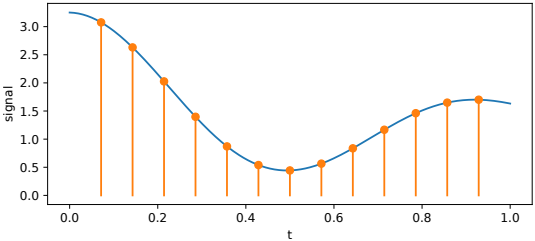

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- **Discrete** $x : \{\ldots, t_1, t_2, t_3, \ldots\} \to \mathbb{R}$,

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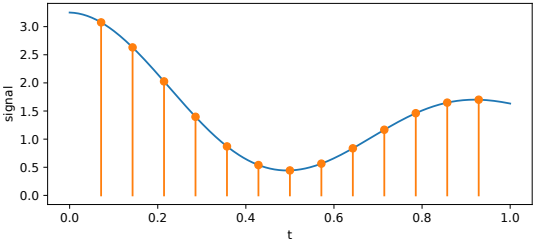
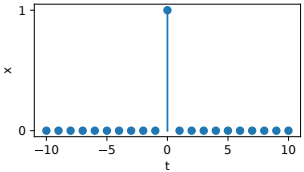


- **Continuous** $x : \mathbb{R} \to \mathbb{R}$,
- **Discrete** $x : \{..., t_1, t_2, t_3, ...\} \to \mathbb{R}$,
- **Discrete** $x : \mathbb{Z} \to \mathbb{R}$ **and** $\delta_t > 0$.

Types of signals

Impulse:

$$\delta[k] = \begin{cases} 1\,, & k = 0\,, \\ 0\,, & k \neq 0\,. \end{cases}$$



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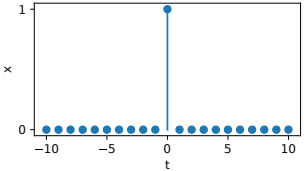
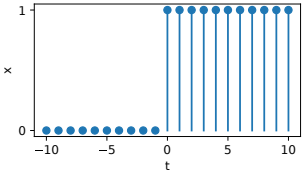


Unit step:

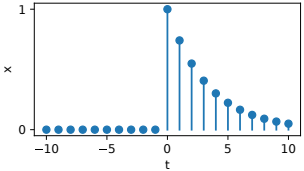
$$u[k] = \begin{cases} 1\,, & k \geq 0\,, \\ 0\,, & k < 0\,. \end{cases}$$



Types of signals (bis)

Exponential decay:

$$x[k] = \begin{cases} e^{-\lambda k}, & k \geq 0, \\ 0, & k < 0. \end{cases}$$



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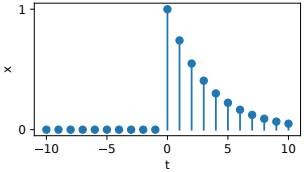
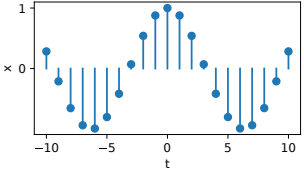


Complex exponential and sinusoid:

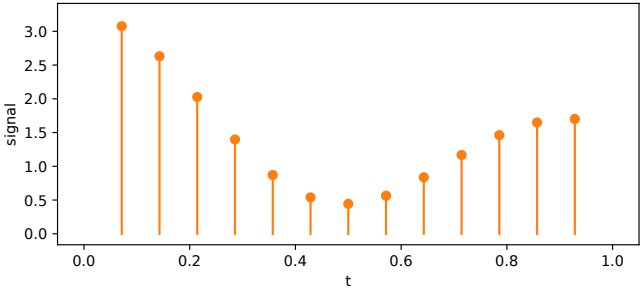
$$x[k] = e^{i\omega k + \phi} = \cos(\omega k + \phi) + i\sin(\omega k + \phi)\,.$$



Reproducing formula

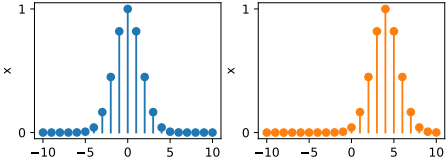
Any discrete signal can be written as a sum of impulses:

$$x[k] = \sum_j x[j]\delta(k-j)\,.$$



Operations

The signal $x[k-n]$ is the signal $x[k]$ **shifted** by n time units to the right.



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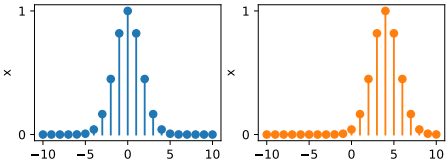
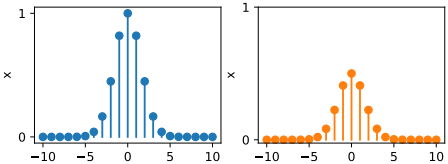
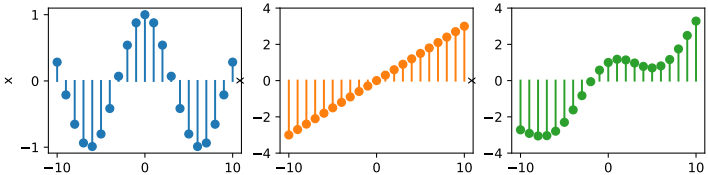


One can rescale a signal by a constant coefficent :



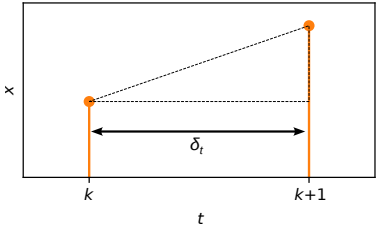
Operations (bis)

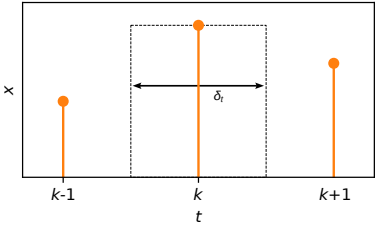
When two signal have the **same sampling times**, we can **add** or **multiply** them.



Operation (ter)

A discrete signal can be **differentiated** or **integrated** :





$$x'[k] = \frac{x[k+1] - x[k]}{\delta_t}\,, \qquad \int_{t_0}^{t_1} x(t)dt = \sum_{k=1}^{T} x[k] \cdot \delta_t\,.$$

Notice that the **impulse** $\delta[k]$ *is the derivative of the* **step** *function* $u[k]$*. And therefore, the step function is the integral of the impulse.*

Energy and power

The **energy** of a signal $x[k]$ is defined as

$$E(x) = \sum_{k=1}^{T} (x[k])^2 \delta_t \,.$$

It matches the notion *energy dissipated through a resistor*, or via any other medium, e.g. a medical device.

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The **power** of a signal is its average energy over a time interval (possibly infinite),

$$P(x) = \frac{1}{T\delta_t} \sum_{k=1}^{T} (x[k])^2 \delta_t = \frac{1}{T} \sum_{k=1}^{T} (x[k])^2 \,.$$

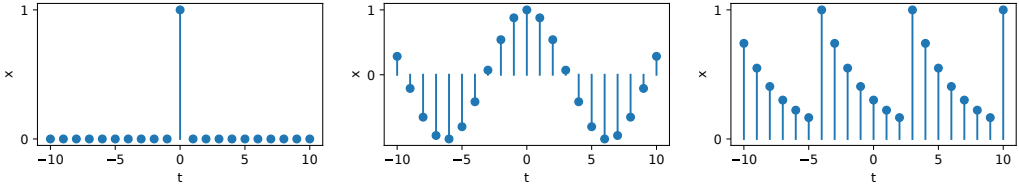
Types of signals

Finite Nonzero over a finite interval of time,

Infinite Nonzero over an infinite interval of time (towards $-\infty$, $+\infty$, or both),

Periodic There exists T such that $x[k+T] = x[k]$ for all k,

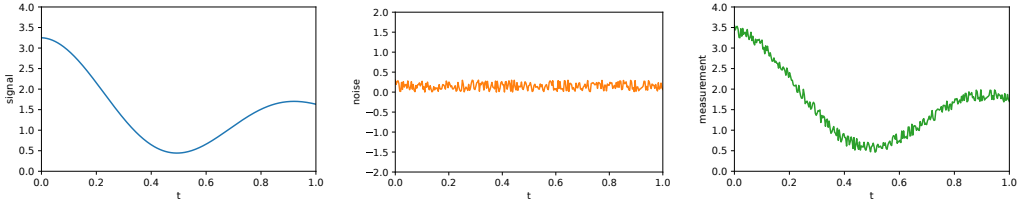
Periodic extension Repeating a finite signal towards ∞.



What is noise?

Any physical system is subject to small disturbances, leading to inaccuracies in the measurements.

The **noise** of a signal is the **discrepancy between the measured value and the real value** of the signal.



A **random variable** X is a function $X : S \to \mathbb{R}$ where:

- S is the sample space: the set of all possible outcomes
- $\mathbb{R}$ set of real numbers, numerical value assigned to the outcome

Example:

Outcome	Value of X
{Heads, Heads}	2
{Heads, Tails}	1
{Tails, Heads}	1
{Tails, Tails}	0

Noise characteristics

Formally, the **noise** at time step k is a **random variable** $\xi[k]$.

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Mostly characterized by its **expectation** and **variance**

$$\mathbb{E}(\xi[k]) = \int_{-\infty}^{+\infty} x p_\xi(x) dx \,, \qquad \mathrm{var}(\xi[k]) = \int_{-\infty}^{+\infty} (x - \mathbb{E}(\xi))^2 \, p_\xi(x) dx \,.$$

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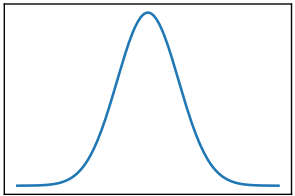
- The expectation is usually assumed to be zero (otherwise you shift all your measurements).
- It is common to assume the noise to be Gaussian.
- The noise is said to be **white** if it is i.i.d. at each time step.



Signal-to-Noise Ratio

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This information is given by the **Signal-to-Noise Ratio (SNR)**, comparing

- the signal's power: $P_{\text{signal}} = \frac{1}{T}\sum_{k=1}^{T}(x[k])^2$,
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$$\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} = \frac{\sum_k (x[k])^2}{\sum_k (\xi[k])^2}\,.$$

A **SNR** larger than 1 means *"more signal than noise"*.

SNR in decibels

Powers can range over many orders of magnitudes.

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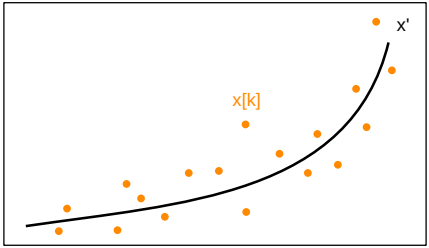
It is common to measure the SNR in **decibels**

$$\mathrm{SNR}_{\mathrm{dB}} = 10\log_{10}(\mathrm{SNR}) = 10\log_{10}(P_{\mathrm{signal}}) - 10\log_{10}(P_{\mathrm{noise}})\,.$$

Then a positive $\mathrm{SNR}_{\mathrm{dB}}$ means *"more signal than noise"*.

Fitting a signal to noisy measurements

I can "see the shape" of the signal despite the noise



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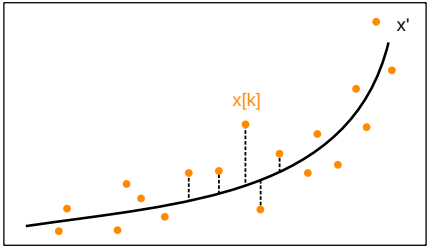

Least square error (LSE):

$$\min_{x'} \sum_{k=1}^{T} (x[k] - x'[k])^2 \,.$$

What else could go wrong?

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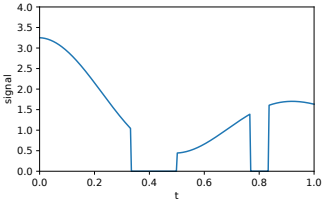
Missing data



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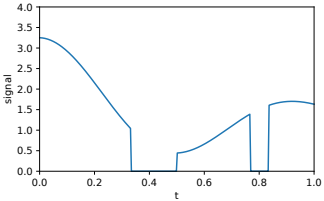
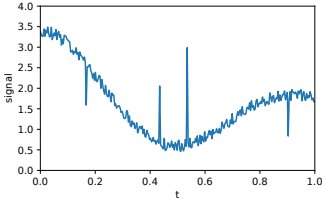


Outlier



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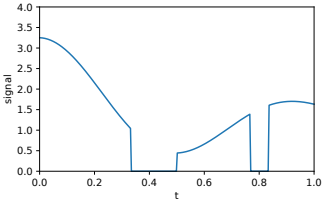

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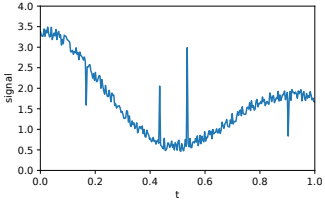
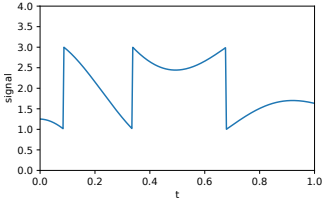


Periodic domain



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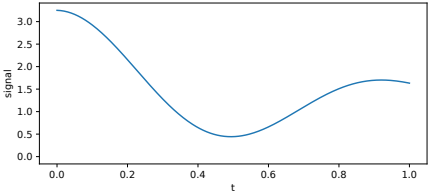
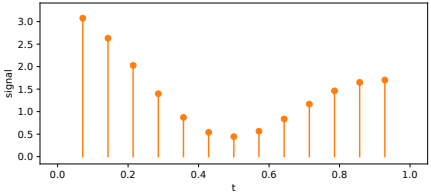
By nature as well, most records are discrete: $x\colon \mathbb{Z} \to \mathbb{Z}$.

Sampling is the process of turning a continuous signal into discrete measurements.





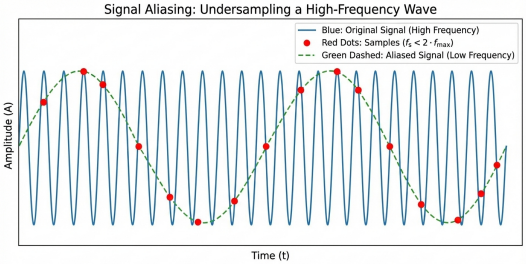
How to sample?

The **sampling rate** corresponds to the number of measurements per seconds.

- **Sampling too fast** is suboptimal. No changes between consecutive measurements.
  Example: Measuring the ambient temperature every second is useless.
- **Sampling too slow** will miss important information.
  Example: Sampling daylight once a day at noon will miss the night...

Undersampling

- Original Signal (High freq): the actual data (e.g., a high-pitched sound, a fast wheel).
- Samples (dots): specific moments where the system records data. The gap between samples is too large.
- Alias (dashed Line): the computer reconstructs the signal by connecting those dots, and it creates a "fake" low-frequency wave.



Undersampling creates a low-frequency alias from a high-frequency signal, violating the Nyquist Criterion.

Aliasing and the Nyquist-Shannon sampling theorem

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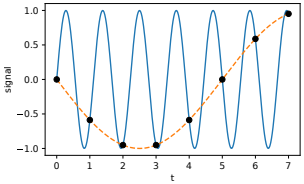
When sampling a periodic signal, sampling too slow may suggest a spurious signal.



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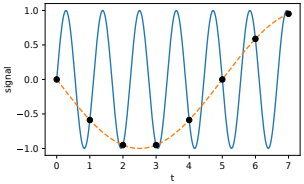

Nyquist-Shannon sampling theorem: *The sampling rate must be at least twice the largest frequency of the signal.*

$$f_s > 2 \cdot f_{max}$$

If you don't meet this condition, the high-frequency information "folds over" into the lower frequencies, creating the distortion we call aliasing.